

LULEA KALLAX, Sweden

WMO: 021860

Lat: 65.543N

Lon: 22.119E

Elev: 20

StdP: 101.09

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-26.3	-23.1	-29.1	0.3	-26.3	-25.6	0.4	-23.1	11.6	0.1	10.3	-0.5	1.4	320	0.684

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.1	23.9	17.2	22.1	16.5	20.8	15.8	18.9	22.2	17.7	20.6	16.6	19.4	3.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.7	12.7	20.5	16.4	11.7	19.3	15.2	10.8	18.0	53.5	22.2	50.1	20.6	46.7	19.5	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	8.3	7.2	-29.7	27.1	3.9	2.5	-32.5	28.9	-34.7	30.4	-36.9	31.8	-39.7	33.6		
(5)			-29.8	19.8	3.8	1.4	-32.6	20.7	-34.8	21.5	-37.0	22.3	-39.8	23.3	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.7	-8.9	-9.3	-4.7	0.9	6.9	12.8	16.3	14.5	9.5	3.0	-2.8	-6.5	(6)	
	DBStd	10.10	7.11	7.42	5.23	3.37	3.73	2.98	2.73	2.92	3.29	4.06	5.42	6.86	(7)	
	HDD10.0	3137	585	542	456	273	111	8	0	2	48	216	384	513	(8)	
	HDD18.3	5722	843	775	714	523	354	168	76	124	265	474	634	771	(9)	
	CDD10.0	475	0	0	0	0	15	91	194	141	33	1	0	0	(10)	
	CDD18.3	18	0	0	0	0	0	1	12	4	0	0	0	0	(11)	
	CDH23.3	75	0	0	0	0	2	9	55	9	0	0	0	0	(12)	
	CDH26.7	8	0	0	0	0	0	1	6	1	0	0	0	0	(13)	
(14) Wind	WSAvg	3.2	3.1	3.1	3.3	3.1	3.3	3.4	3.2	3.1	3.3	3.2	3.1	3.0	(14)	
(15) Precipitation	PrecAvg	532	39	32	34	31	31	40	52	67	55	52	55	45	(15)	
	PrecMax	681	89	61	80	65	65	96	150	140	103	112	98	149	(16)	
	PrecMin	380	4	6	3	11	4	4	10	19	14	12	18	12	(17)	
	PrecStd	77	19	15	23	17	16	25	31	34	21	26	23	32	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.0	7.0	9.0	13.1	21.7	24.4	27.2	24.5	19.0	12.9	8.8	6.2	(19)	
		MCWB	4.0	4.4	4.9	7.5	14.3	15.2	18.7	17.9	13.6	8.7	6.3	4.3	(20)	
	2%	DB	3.1	4.4	6.5	10.3	18.2	21.8	24.8	22.5	17.1	10.9	6.6	4.2	(21)	
		MCWB	1.7	2.1	3.1	5.6	11.8	14.6	18.1	17.2	13.1	9.3	5.7	2.8	(22)	
	5%	DB	1.4	1.9	4.5	8.2	15.8	19.9	23.0	21.0	15.8	9.9	5.2	2.9	(23)	
		MCWB	0.7	0.6	1.7	4.3	10.2	13.7	17.0	16.5	12.7	8.8	4.6	2.1	(24)	
	10%	DB	0.2	0.5	2.3	6.3	13.3	18.2	21.5	19.6	14.6	8.8	4.0	1.9	(25)	
		MCWB	-0.2	-0.1	0.4	3.2	8.6	12.9	16.5	15.8	12.4	7.7	3.3	1.3	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.1	4.4	5.2	8.1	14.7	17.9	20.6	20.1	15.4	10.9	7.0	4.5	(27)	
		MCDB	5.8	6.8	8.3	12.6	20.2	21.2	24.3	22.9	17.2	11.4	7.6	5.7	(28)	
	2%	WB	2.0	2.3	3.4	6.1	12.6	15.9	19.4	18.5	14.3	9.8	5.9	3.1	(29)	
		MCDB	2.9	4.0	6.1	9.5	17.4	19.2	23.2	21.0	15.9	10.3	6.5	3.8	(30)	
	5%	WB	0.8	0.8	1.8	4.8	10.6	14.8	18.4	17.3	13.6	8.8	4.7	2.2	(31)	
		MCDB	1.3	1.4	3.8	7.7	14.7	18.3	21.7	19.8	15.0	9.6	5.1	2.7	(32)	
	10%	WB	-0.1	0.0	0.7	3.5	9.1	13.7	17.3	16.1	12.9	7.6	3.4	1.2	(33)	
		MCDB	0.3	0.5	1.9	5.9	12.7	17.3	20.3	18.7	14.2	8.6	3.9	1.7	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	7.9	8.4	7.7	8.1	7.3	7.1	7.0	6.8	5.7	5.5	6.6	(35)	
		MCDBR	7.2	9.1	9.5	10.5	11.5	9.8	9.5	8.6	8.0	6.3	5.3	5.8	(36)	
	5% WB	MCWBR	6.6	7.7	7.0	6.8	6.4	4.9	4.6	4.7	5.1	4.9	4.9	5.3	(37)	
		MCWBR	6.3	7.4	8.6	10.1	10.8	8.4	7.8	7.2	6.2	5.2	4.8	5.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		N/A	0.272	0.301	0.335	0.343	0.347	0.362	0.347	0.319	0.298	0.340	0.349	(40)	
		taud	N/A	2.135	2.208	2.282	2.442	2.490	2.468	2.500	2.554	2.382	2.610	2.633	(41)	
	Ebn at Noon		N/A	596	763	829	864	867	839	817	767	584	264	95	(42)	
		Edh at Noon	N/A	56	84	100	95	93	93	82	62	45	19	10	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.09	0.66	2.20	4.01	5.05	5.63	5.19	3.82	2.11	0.78	0.15	0.02	(44)	
	RadStd		0.01	0.11	0.20	0.23	0.46	0.44	0.46	0.34	0.28	0.14	0.03	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (42 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page